
Explicit vs Implicit World Models for Decision Making: A Comparison of Generative and Task-Centric Latent Representations

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Abstract

Model-based reinforcement learning agents differ fundamentally in the objective used to shape their latent representation: generative agents reconstruct observations, while task-centric agents optimise consistency and temporal-difference losses. We implement and compare a DreamerV3-style Recurrent State-Space Model (RSSM) and a TD-MPC2-style implicit world model on the `ManiSkill2 LiftCube-v0` manipulation task, and characterise their latent spaces via linear probing, effective dimensionality analysis, reward predictability probing, and Canonical Correlation Analysis (CCA). The implicit agent achieves substantially higher task performance (mean return 90.7 vs. 8.9), while the explicit agent produces latents that are more faithfully decodable to raw observation coordinates (median $R^2 = 0.9998$ vs. 0.934). A reward predictability probe reveals the mechanistic explanation: the implicit agent linearises the reward landscape in its latent space (gain +1.75 over the observation baseline), whereas the explicit agent’s latent is effectively a compressed copy of the observation, which does not provide additional reward signal. CCA reveals that both agents discover the same primary axes of state variation despite different architectures and losses, with top canonical correlations above 0.97.

1. Introduction

Model-based reinforcement learning (MBRL) agents learn a compressed world model to support planning and policy optimisation without requiring a simulator at inference time (Hafner et al., 2020; Hansen et al., 2022). Two broad

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representational families have emerged with different inductive biases.

Explicit world models (Hafner et al., 2021; 2023) learn a generative model that can reconstruct observations. The latent state must preserve enough information to regenerate what was seen, creating direct gradient pressure to encode all observation dimensions. Policy learning then occurs entirely in the imagined latent space via actor-critic methods (Schulman et al., 2016).

Implicit world models (Hansen et al., 2022; 2024) forgo reconstruction entirely. The latent is trained via a consistency loss (predicted next-state should match the encoder’s output) and temporal-difference (TD) losses on rewards and Q-values. At decision time, planning is performed in the latent space via MPPI (Williams et al., 2017).

The central question examined here is: *Does generative reconstruction pressure produce latent representations that are more linearly decodable in terms of physical state coordinates, and does that difference in representational structure translate to a difference in control performance?*

We implement both families in a unified training harness, run them on the `LiftCube-v0` manipulation task from `ManiSkill2` (Gu et al., 2023), and conduct five latent-space analyses designed to reveal structure invisible in performance curves alone.

2. Background

The foundational vision. Ha & Schmidhuber (2018) demonstrated that an agent can learn a compressed spatial and temporal representation of its environment in an unsupervised manner, effectively building a “mental model” that allows it to simulate future trajectories. This established the core idea that a complex environment can be distilled into a compact representation, making downstream policy learning significantly more efficient.

Explicit world models. The Dreamer series (Hafner et al., 2020; 2021; 2023) builds on this by using the Recurrent State-Space Model (RSSM) to learn an environment model that predicts future observations, rewards, and terminal signals. DreamerV2 demonstrated that behaviors can be

learned purely through “latent imagination” — propagating gradients through imagined trajectories in latent space. DreamerV3 extended this into a general-purpose algorithm capable of mastering diverse domains with fixed hyperparameters.

Implicit world models. TD-MPC2 (Hansen et al., 2024) shifts focus from reconstruction to prediction of task-relevant information. Rather than imagining pixel-perfect futures, it uses Temporal Difference (TD) learning and Model Predictive Control (MPC) to shape a latent space guided by value-driven objectives, achieving high sample efficiency across continuous-control tasks without the computational overhead of generating future observations.

The central tension. The trade-off under investigation is between *generative fidelity* and *task-specific utility*. Explicit models offer a rich interpretable internal world that may benefit complex planning, but risk wasting capacity on task-irrelevant details. Implicit models are more streamlined, encoding only what is necessary to maximise value. Our experiments quantify these trade-offs and investigate what each architecture encodes about the world it inhabits.

3. Agents and Architecture

3.1. Explicit Agent (RSSM / DreamerV3-style)

The explicit agent learns a Recurrent State-Space Model (RSSM) (Hafner et al., 2019) whose latent state comprises two parts.

Deterministic state $h_t \in \mathbb{R}^{256}$ is a GRU hidden state:

$$h_t = \text{GRU}(h_{t-1}, [z_{t-1}; a_{t-1}]). \quad (1)$$

Stochastic state z_t is sampled from a discrete categorical posterior with 8 variables $\times$ 8 bins, yielding a 64-dimensional one-hot vector:

$$z_t \sim \text{Cat}(\text{MLP}(h_t, \text{embed}_t)). \quad (2)$$

The full feature vector fed to the policy is $[h_t; z_t^{\text{flat}}] \in \mathbb{R}^{320}$. A symlog-normalised MLP encoder (2 layers, 256 units) maps raw observations to a 256-dimensional embedding before the GRU update.

The world model is trained jointly with three prediction heads:

KL divergence between the learned posterior and a prior is minimised with free bits ($\text{kl_free} = 1.0$, $\lambda_{\text{dyn}} = 0.5$, $\lambda_{\text{rep}} = 0.1$).

An **actor-critic** (*ImagBehavior*) is trained entirely in imagination: the actor rolls out $H = 10$ steps in latent space and a λ -return critic ($\lambda = 0.95$, $\gamma = 0.997$) with a slow EMA target provides value estimates. The world model

Table 1. Explicit agent prediction heads.

HEAD	OUTPUT	LOSS
DECODER	RECONSTRUCTED OBS (42-DIM)	SYMLOG MSE
REWARD	255-BIN SYMLOG-DISCRETE	CROSS-ENTROPY
CONTINUATION	BERNOULLI (1-DONE)	BINARY CE

is updated every `train_ratio=64` environment steps (3,125 gradient updates over 200k steps); batch size is 16 sequences $\times$ 64 steps.

3.2. Implicit Agent (TD-MPC2-style)

The implicit agent learns a 64-dimensional continuous latent space without any decoder (Hansen et al., 2024). The encoder uses SimNorm activations (simplicial normalisation that maps each group of 8 latent units onto the probability simplex, preventing representational collapse):

$$z_t = \text{SimNorm}(\text{MLP}(\text{symlog}(\text{obs}_t))) \in \mathbb{R}^{64}. \quad (3)$$

A latent dynamics model predicts $z_{t+1} = \text{SimNorm}(\text{MLP}([z_t; a_t]))$. A 2-ensemble Q-network maps (z_t, a_t) to 101-bin two-hot encoded Q-values.

Training losses (batch: 256 transitions, horizon $H = 5$):

Table 2. Implicit agent training objectives.

LOSS	DESCRIPTION	WEIGHT
CONSISTENCY	$\ \text{dyn}(z_t, a_t) - \text{sg}(z_{t+1})\ ^2$	20.0
REWARD	SOFT CROSS-ENTROPY (TWO-HOT)	0.1
VALUE	TD LOSS (TWO-HOT, MIN-Q TARGET)	0.1

Action selection uses MPPI/CEM planning (Williams et al., 2017; Rubinstein, 1999): 4 CEM iterations $\times$ 64 sampled trajectories $\times$ $H = 5$ horizon in latent space, warm-started from the previous step’s plan. One gradient update is performed every 2 environment steps after 1,000 seed steps ($\approx 100\text{k}$ updates over 200k steps).

3.3. Shared Infrastructure

Both agents are evaluated on ManiSkill2 (Gu et al., 2023) `LiftCube-v0` (`obs_dim=42`, `act_dim=8`, dense reward, 200 steps per episode, action repeat=2). Raw observations are stored in a replay buffer of capacity 200k transitions; symlog normalisation is applied inside each agent’s encoder. Checkpoints are retained at evenly-spaced steps across training to support temporal evolution analysis.

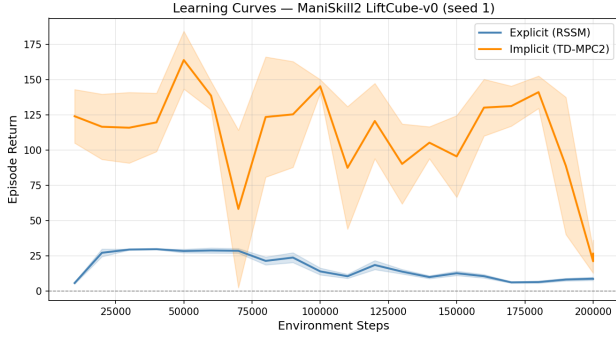


Figure 1. Learning curves on LiftCube-v0 (200k environment steps, seed 1). Shaded regions show ± 1 std across evaluation episodes (10 per eval).

Table 3. Performance summary (last 50k environment steps, seed 1).

AGENT	MEAN RETURN	STD	PEAK
EXPLICIT (RSSM)	8.9	2.1	29.8
IMPLICIT (TD-MPC2)	90.7	45.8	163.8

4. Experimental Comparison

4.1. Learning Curves and Performance

Figure 1 and Table 3 show that the implicit agent substantially outperforms the explicit agent. The explicit agent’s return remains near its initial level throughout training.

4.2. Failure Mode Analysis

Explicit agent — training ratio sensitivity. The `train_ratio` parameter controls world model update frequency. At the DreamerV3 default of 512 (tuned for 2M-step DMControl budgets), the explicit agent performed only 390 world model gradient updates over 200k steps — insufficient to learn reliable manipulation dynamics. Reducing `train_ratio` to 64 raised the update count to 3,125 and produced markedly faster early learning (return ≈ 30 at step 40k). This sensitivity is itself a meaningful failure mode: the explicit agent requires careful tuning of the gradient-to-environment-step ratio in ways the implicit agent does not.

Implicit agent — instability. The implicit agent achieves high returns but with large variance (std = 45.8). This pattern — rapid initial learning followed by oscillation — is consistent with the MPPI planner exploiting overfit Q-value estimates as the buffer composition shifts.

Dataset limitation. Results are reported for a single environment and a single seed, constrained by compute budget.

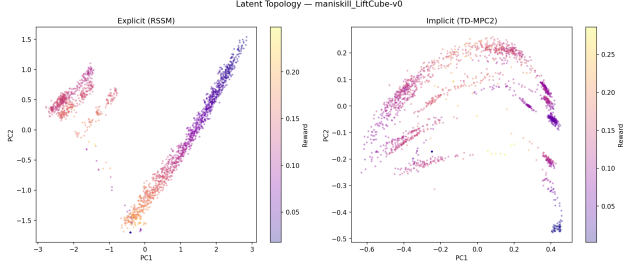


Figure 2. 2D PCA projections coloured by step reward. The explicit latent forms a tighter, more structured manifold; the implicit latent is more diffuse.

Patterns described should be treated as observations from one run, not statistically established claims.

5. Latent Space Analysis

This section characterises what each agent’s latent space encodes, using analyses chosen to reveal structure that cannot be read from performance curves.

5.1. Linear Probing

A Ridge regression ($\alpha = 1.0$, 5-fold CV) was trained to predict each of the 42 observation coordinates from the frozen latent vector.

Table 4. Linear probing R^2 of 42 observation coordinates.

METRIC	EXPLICIT (RSSM)	IMPLICIT (TD-MPC2)
MEDIAN R^2	0.9998	0.934
MEAN R^2	0.961	0.686

The explicit agent’s latent is nearly perfectly decodable to observation coordinates (median $R^2 \approx 1.0$), as expected: the decoder loss creates direct gradient pressure to preserve all observation information. The implicit agent’s latent is less faithfully invertible; coordinates 9–17 (end-effector velocity estimates) show $R^2 \approx 0.23$.

High linear decodability means the latent *contains* the observation information — it does not imply better task performance, as the following analyses show.

5.2. Effective Dimensionality

PCA was applied to 2,000 standardised latent vectors to measure how many principal components are actually used (Table 5).

Both agents collapse their nominal latent dimensions into

Table 5. Effective dimensionality via PCA.

METRIC	EXPLICIT	IMPLICIT
NOMINAL LATENT DIM	256	64
DIMS FOR 90% VAR	5	7
DIMS FOR 95% VAR	6	9
DIMS FOR 99% VAR	11	15
PARTICIPATION RATIO	3.24/256	3.98/64
TOP-5 VAR SHARE	93.6%	86.2%

very few effective dimensions. The explicit agent uses only 6 effective dimensions at the 95% threshold despite a 256-dimensional nominal space, with a single component explaining 49.5% of variance. The implicit agent uses 9 effective dimensions (64-dimensional nominal), top component 43.2%. Both are far smaller than the 42-dimensional observation space, and similar in absolute terms despite a $4\times$ difference in nominal latent size.

5.3. Reward Predictability

A Ridge regression was trained to predict the next-step reward from three feature sets: raw observations, latent alone, and their concatenation.

 Table 6. Reward predictability probe (R^2).

FEATURE	EXPLICIT	IMPLICIT
OBSERVATION ONLY	0.983	-0.910
LATENT ONLY	0.995	0.842
LATENT + OBS	0.995	0.478
GAIN (LATENT-OBS)	+0.012	+1.752

This analysis most clearly distinguishes the two representations. For the **explicit agent**, raw observations already predict next-step reward with $R^2 = 0.983$; the latent adds only 0.012 over that baseline. The explicit latent is essentially a compressed but faithful copy of the observation, and the observation itself is nearly sufficient for reward prediction.

For the **implicit agent**, raw observations achieve $R^2 = -0.91$: the relationship between raw observation coordinates and reward is nonlinear and cannot be captured by a linear probe. The latent, however, achieves $R^2 = 0.842$. The implicit encoder has *linearised* the reward-relevant structure: its latent is a coordinate system in which the reward landscape is approximately linear, even though the raw observation space is not.

This explains why the explicit agent’s higher linear probing R^2 does not translate to better control performance. The explicit agent learns a representation that faithfully preserves observation structure; the implicit agent learns one that reorganises observations to make reward predictable.

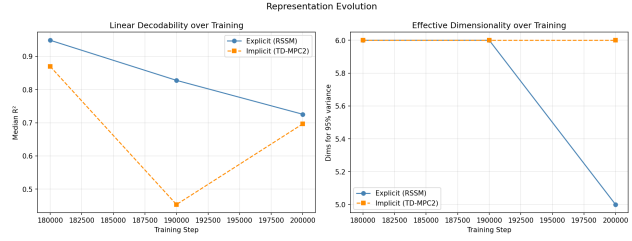


Figure 3. Representation evolution over the last 20k training steps (checkpoints at 180k, 190k, 200k). Left: median linear probing R^2 . Right: effective dimensions at the 95% variance threshold.

5.4. CCA Alignment

CCA was applied to latent sequences produced by both agents on identical observation trajectories (2,000 steps, same environment seed).

Table 7. Canonical Correlation Analysis (CCA) between agent latents.

METRIC	VALUE
MEAN CANONICAL CORR.	0.896
TOP-3 CANONICAL CORRS.	0.997, 0.981, 0.969

The first three canonical dimensions have near-perfect correlation (> 0.97). Both agents discover essentially the same primary axes of state variation, despite different loss functions, architectures, and nominal latent dimensions. This is consistent with the effective dimensionality results: if both agents reduce the world to 6–9 effective dimensions and the task constrains which directions matter, there is limited room for representations to diverge in their primary axes. The mean drops to 0.896 averaged over 10 components because later components capture more agent-specific structure.

5.5. Representation Evolution

Figure 3 shows that neither representation has stabilised by step 200k. The explicit agent’s median R^2 declines from 0.949 (step 180k) to 0.725 (200k), with effective dimensionality decreasing from 6 to 5. The implicit agent’s R^2 oscillates between 0.453 and 0.870.

The explicit agent’s declining R^2 in late training is consistent with the GRU’s deterministic state gradually forgetting early observation structure as the world model receives insufficient updates to consolidate it. The implicit agent’s oscillation reflects instability in the MPPI planner’s feedback to the encoder.

6. Summary and Hypothesis Assessment

The hypothesis was that generative reconstruction pressure produces representations with higher linear decodability, and that this physical grounding translates to better control performance.

Supported: Reconstruction pressure does produce latents that are more linearly decodable in terms of raw observation coordinates (explicit median $R^2 = 0.9998$ vs. implicit 0.934). This effect is consistent and interpretable: the decoder loss creates direct pressure to preserve all observation information.

Not supported: Higher observation-decodability does not translate to better control. The implicit agent performed approximately $10\times$ better (mean return 90.7 vs. 8.9). The reward predictability probe provides a mechanistic account: in this environment, raw observation coordinates do not linearly predict reward ($R^2 = -0.91$), so a representation that faithfully copies the observation does not help the actor. The implicit agent’s consistency and TD losses produce a representation in which the reward landscape is linearised ($R^2 = 0.842$ from latent alone), supporting better planning and policy learning.

Additional finding: Both representations are far more concentrated than their nominal dimensionality suggests (6 and 9 effective dims respectively). The high CCA alignment (mean 0.896, top component 0.997) indicates both agents discover the same primary structure. The latent geometry differences that matter most for control — the reward-predictive structure — are not visible in the primary canonical dimensions but in how agents use their remaining degrees of freedom.

Confound: The performance comparison is complicated by the explicit agent’s sensitivity to `train_ratio`. The experiment checkpoint was produced with `train_ratio=512`, giving only 390 world model gradient updates over 200k steps. With `train_ratio=64`, the explicit agent showed markedly faster early learning (return ≈ 30 at step 40k). A fair asymptotic comparison requires either longer training (500k–1M steps) or a matched gradient-update budget, and results should be interpreted with this caveat.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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